

IMU SIGN LANGUAGE TRANSLATOR

Arduino UNO • 5× MPU6050 • TCA9548A • HC-05

Project Documentation

GitHub Repository: [Girdhari-Chaturvedi/Imu-Sign-Language-Translator](https://github.com/Girdhari-Chaturvedi/Imu-Sign-Language-Translator)

1. Project Overview

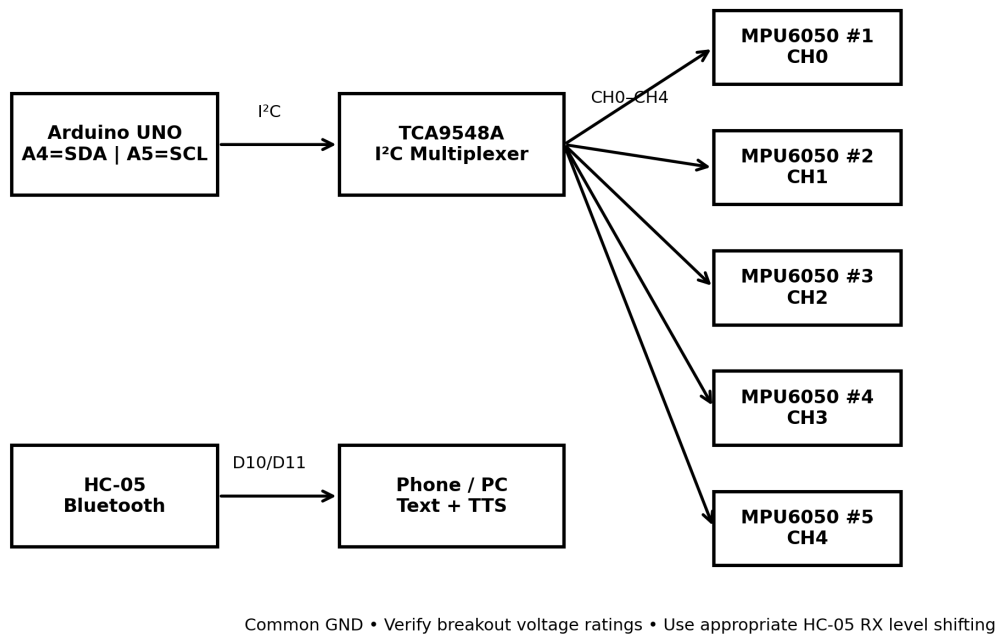
A wearable motion-based sign/gesture translator using five MPU6050 sensors connected through a TCA9548A I²C multiplexer. Arduino UNO reads the sensors, extracts motion features, applies a configurable prototype gesture vocabulary, and sends structured data through USB and HC-05 Bluetooth.

2. Main Components

Component	Qty	Purpose
Arduino UNO	1	Main controller
MPU6050 / GY-521	5	Motion sensing
TCA9548A	1	I ² C multiplexing
HC-05	1	Bluetooth serial
Breadboard	1+	Prototype wiring
Jumper wires	As required	Connections
USB cable	1	Programming / USB serial
Resistors / level shifting	As required	HC-05 RX protection

3. Circuit Diagram

Five MPU6050 boards use separate TCA9548A channels, allowing identical sensor addresses to coexist.



4. Pin Connections

Arduino UNO	Connection
A4 / SDA	TCA9548A SDA
A5 / SCL	TCA9548A SCL
GND	Common ground
D10	HC-05 TXD
D11	HC-05 RXD through appropriate divider / level shifting

TCA9548A Channel Mapping

Sensor	TCA Channel	Typical MPU6050 Address
MPU6050 #1	0	0x68
MPU6050 #2	1	0x68
MPU6050 #3	2	0x68
MPU6050 #4	3	0x68
MPU6050 #5	4	0x68

5. Working Logic

1. Initialize Arduino I²C and TCA9548A. 2. Select each TCA channel. 3. Initialize and read its MPU6050. 4. Convert raw acceleration and gyro readings. 5. Calculate pitch and roll. 6. Calibrate neutral orientation at startup. 7. Combine motion features from connected sensors. 8. Evaluate prototype gesture rules. 9. Send DATA telemetry and SIGN messages through USB and HC-05.

6. Communication Format

Telemetry: DATA,S1,AX=0.120,AY=-0.940,AZ=0.310,GX=2.0,GY=-1.0,GZ=0.5,P=1.2,R=-70.3

Gesture: SIGN:HELLO

7. Prototype Gesture Vocabulary

YES, NO, HELP, THANK YOU, HELLO, WATER, FOOD, BYE, COME HERE, GO, STOP, WAIT, GOOD MORNING, GOOD NIGHT, EMERGENCY, I NEED HELP, PLEASE, SORRY, OK.

8. Software

Arduino firmware is stored under `firmware/arduino_5x_mpu6050/`. The optional Windows desktop application under `software/desktop_usb_voice/` reads USB serial data and uses `pyttsx3` for text-to-speech.

9. Important Limitation

The included gesture vocabulary is a threshold-based prototype. It should not be presented as complete recognition of every sign-language word. Robust recognition requires consistent sensor placement, calibration, labeled time-series data, training, and validation.

10. Future Development

Create a five-sensor data logger, build a labeled dataset, add orientation normalization, train an ML classifier with confidence scoring, add an Android Bluetooth/TTS app, and design a wearable/3D-printed sensor mount.